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ITAO7103 Assignment 1 — Predicting Star Performance in Collaboration Networks

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1. Introduction

This report examines the drivers of high performance in collaboration networks by testing three competing theoretical perspectives from the statistical learning and network science literature.

First, the elite learning perspective (Liu et al., 2018) suggests that exposure to high-performing collaborators enhances individual productivity through knowledge transfer and skill acquisition. Second, structural holes theory (Burt, 2004) argues that individuals occupying brokerage positions in less constrained networks gain access to diverse, non-redundant information, leading to higher performance. Third, the technological breadth perspective (Fleming, 2001) proposes that performance is driven by exposure to diverse knowledge domains rather than repeated or cohesive collaborations.

Using a large-scale collaboration network dataset, this study evaluates how elite exposure (e.g. Num_Elite_Collaborators), network structure (e.g. Network_Constraint, Effective_Size), and collaboration diversity jointly influence productivity (Total_Impact) and high-performer status (Is_HighPerformer).

To address these questions, the analysis applies a structured statistical learning pipeline: exploratory data analysis, linear and logistic regression, LASSO-based feature selection, and classification models (logistic regression and LDA). These methods allow both inference (identifying significant predictors) and prediction (evaluating classification performance), providing a comprehensive assessment of which theoretical mechanisms are most strongly supported by the data.

2. Methodology

A complete-case analysis was applied after inspecting the missing data structure. The missingness pattern was not random; specifically, elite-related variables (Cohesion_Elite, Expertise_Sim_Elite, Repeated_Elite) exhibited extensive missingness due to their conditional definition — these metrics are only computed for individuals with at least one elite collaborator. As a result, their missingness reflects structural absence of elite ties rather than data quality issues.

Similarly, partial missingness in network structure variables (Network_Constraint, Effective_Size, Network_Efficiency) is likely attributable to network boundary effects and isolated or sparsely connected nodes where structural measures cannot be reliably computed.

Rather than imputing these values, which would introduce strong model-based assumptions and potentially distort network structure, a listwise deletion strategy was adopted across all variables to ensure a consistent analytic sample. This ensures that all observations used in modelling have a fully observed feature vector, which is essential for methods such as LASSO and logistic regression that do not natively handle missing data.

Although this approach reduces sample size, it avoids introducing bias from inappropriate imputation in non-random missingness contexts (MNAR), particularly for network-derived features where missingness carries substantive meaning.

```
DATA_PATH <- "D:/Sanjai/Assignments/advanced analytics dataset.xlsx"
```

```
library(readxl)
df <- read_excel(DATA_PATH)

df <- df %>%
  mutate(
    Is_HighPerformer = as.factor(Is_HighPerformer),
    Log_Impact = log(Total_Impact + 1)
  )

colSums(is.na(df))
```

```
##           AID           Is_HighPerformer           Total_Impact
##           0              0              0
##   Total_Projects   Career_Length           Num_Topics
##           0              0              0
##   Num_Domains     Num_Collaborators   Num_Elite_Collaborators
##           0              0              0
##   Cohesion_Elite   Expertise_Sim_Elite   Repeated_Elite
##   101666           101666           101666
##   Projects_With_Elite   Num_Regular_Collaborators   Cohesion_Regular
##           0              0              5881
##   Expertise_Sim_Regular   Repeated_Regular   Network_Constraint
##           5881           5881           15206
##   Effective_Size     Network_Efficiency   Log_Impact
##           5820           5820              0
```

```
df <- na.omit(df)
colSums(is.na(df))
```

```
##           AID           Is_HighPerformer           Total_Impact
##           0              0              0
##   Total_Projects   Career_Length           Num_Topics
##           0              0              0
##   Num_Domains     Num_Collaborators   Num_Elite_Collaborators
##           0              0              0
##   Cohesion_Elite   Expertise_Sim_Elite   Repeated_Elite
##           0              0              0
##   Projects_With_Elite   Num_Regular_Collaborators   Cohesion_Regular
##           0              0              0
##   Expertise_Sim_Regular   Repeated_Regular   Network_Constraint
##           0              0              0
##   Effective_Size     Network_Efficiency   Log_Impact
##           0              0              0
```

```
cat("Rows:", nrow(df), "Columns:", ncol(df))
```

```
## Rows: 7762 Columns: 21
```

3. Part 1: Exploratory Data Analysis

3.1 Summary Statistics

```
vars <- c("Total_Impact", "Total_Projects", "Career_Length",
         "Num_Elite_Collaborators", "Num_Regular_Collaborators",
         "Network_Constraint", "Effective_Size")

summary_table <- df %>%
  select(all_of(vars)) %>%
  summary()

kable(summary_table) %>% kable_styling()
```

Total_Impact	Total_Projects	Career_Length	Num_Elite_Collaborators	Num_Regular_Collaborators	Network_Constraint	Effective_Size
Min. : 0.0	Min. : 1.000	Min. : 1.00	Min. : 1.000	Min. : 1.00	Min. :0.004382	Min. : 1.00
1st Qu.: 50.0	1st Qu.: 1.000	1st Qu.: 1.00	1st Qu.: 1.000	1st Qu.: 5.00	1st Qu.:0.120006	1st Qu.: 1.00
Median : 153.0	Median : 2.000	Median : 3.00	Median : 1.000	Median : 10.00	Median :0.261112	Median : 4.75
Mean : 523.3	Mean : 4.858	Mean : 4.43	Mean : 2.205	Mean : 17.12	Mean :0.331156	Mean : 12.29
3rd Qu.: 496.0	3rd Qu.: 5.000	3rd Qu.: 8.00	3rd Qu.: 2.000	3rd Qu.: 21.00	3rd Qu.:0.492711	3rd Qu.: 14.35
Max. :24809.0	Max. :470.000	Max. :11.00	Max. :29.000	Max. :277.00	Max. :1.125000	Max. :292.24

Interpretation: The summary statistics indicate substantial dispersion across key performance and network variables, suggesting heterogeneity in collaboration structures and output. In particular, Total_Impact is highly right-skewed, which justifies the use of the log transformation to stabilise variance and reduce the influence of extreme values. a standard approach in productivity research (e.g.Robert K. Merton’s work on cumulative advantage).

Variables such as Num_Elite_Collaborators and Total_Projects show wide ranges, implying that exposure to elite collaborators is unevenly distributed. This is consistent with the interpretation that access to high-value networks is non-randomly distributed. Meanwhile, Network_Constraint and Effective_Size capture structural positioning, where higher constraint indicates closed networks and higher effective size reflects brokerage opportunities.

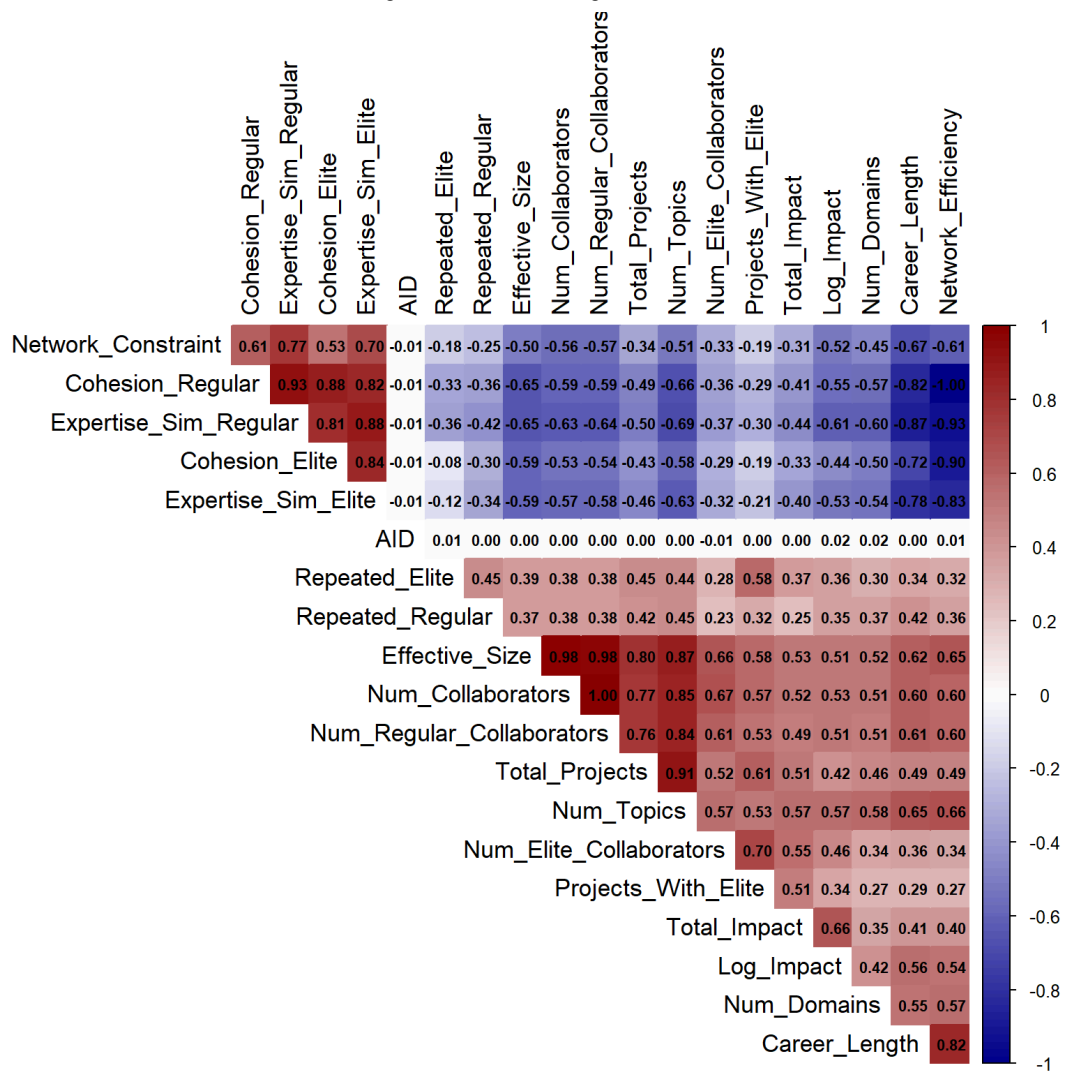
The gap between mean and median in several variables further confirms skewness, reinforcing that the distribution is right-skewed, indicating concentration of high impact among a small subset of individuals.

3.2 Correlation Matrix

```
num_data <- df %>%
  select(where(is.numeric)) %>%
  na.omit()

corr_matrix <- cor(num_data, use = "complete.obs")

corrplot(
  corr_matrix,
  method = "color",
  type = "upper",
  order = "hclust",
  addCoef.col = "black",
  tl.col = "black",
  tl.cex = 1.1,          # increase label size (IMPORTANT)
  number.cex = 0.7,     # correlation numbers bigger
  col = colorRampPalette(c("darkblue", "white", "darkred"))(200),
  diag = FALSE
)
```



Interpretation: The correlation heatmap reveals moderate but not extreme correlations, with no pair exceeding the critical $|r| > 0.7$ threshold. This indicates that severe multicollinearity is not present, which is later confirmed by VIF values. However, conceptually meaningful relationships emerge:

- Career_Length and Total_Projects are moderately correlated as expected, as longer careers allow accumulation of projects.
- Effective_Size and Network_Constraint show an inverse relationship is structurally consistent with Ronald Burt's theory: open networks reduce constraint.
- Num_Elite_Collaborators correlates positively with impact variables. It is consistent with patterns reported in social network analysis literature on knowledge spillovers.
- Even though correlations are not extreme, the presence of moderate clustering among network variables suggests conceptual multicollinearity, meaning variables capture overlapping theoretical constructs (e.g. cohesion vs brokerage). This indicates potential conceptual overlap between network constructs, which may affect coefficient interpretation in multivariate models.

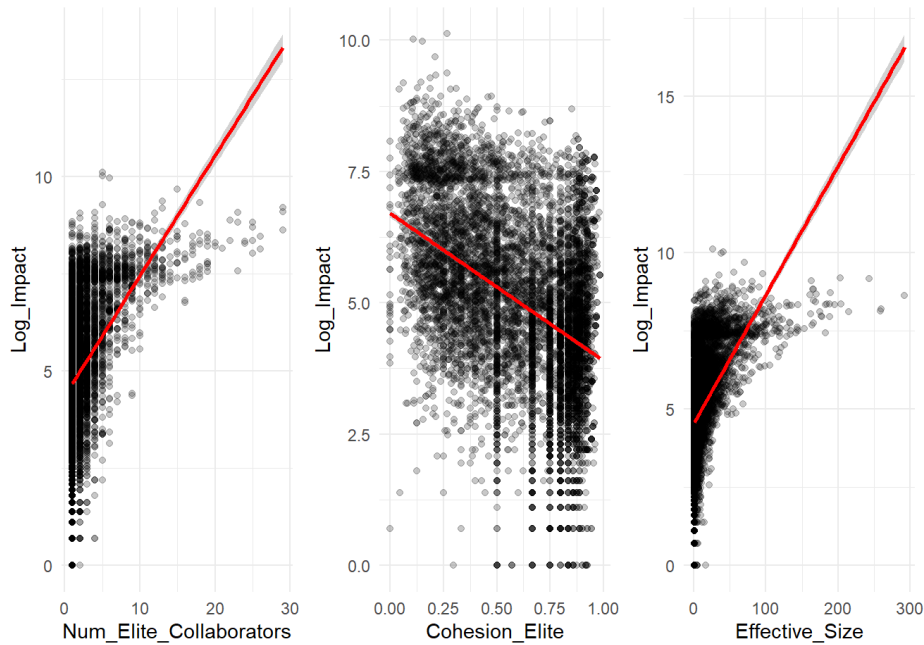
3.3 Scatter Plots

```
p1 <- ggplot(df, aes(Num_Elite_Collaborators, Log_Impact)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", color = "red") +
  theme_minimal()

p2 <- ggplot(df, aes(Cohesion_Elite, Log_Impact)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", color = "red") +
  theme_minimal()

p3 <- ggplot(df, aes(Effective_Size, Log_Impact)) +
  geom_point(alpha = 0.2) +
  geom_smooth(method = "lm", color = "red") +
  theme_minimal()

p1 + p2 + p3
```



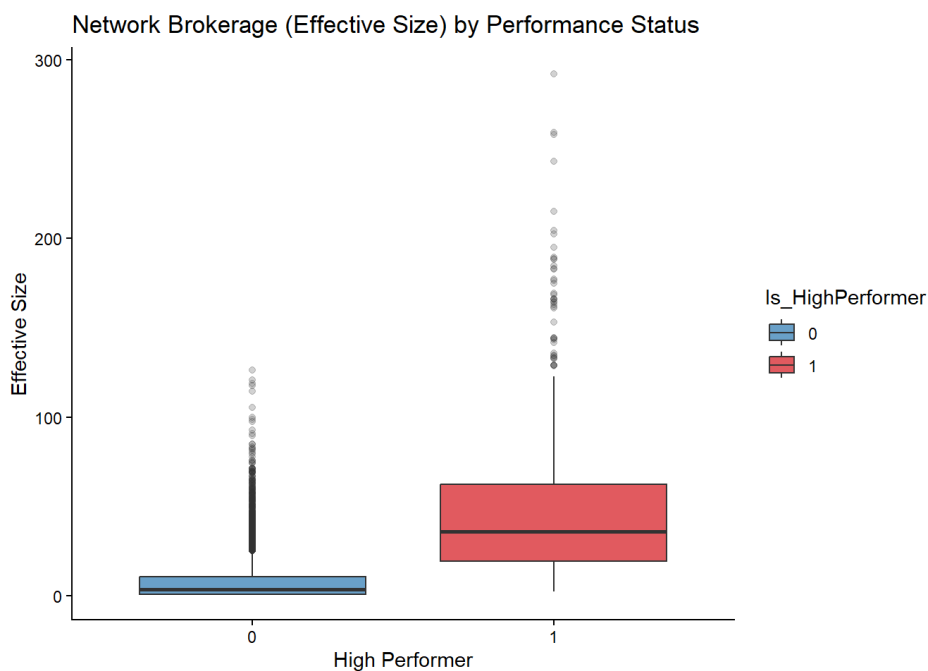
Interpretation: The scatter plots reveal

distinct structural patterns:

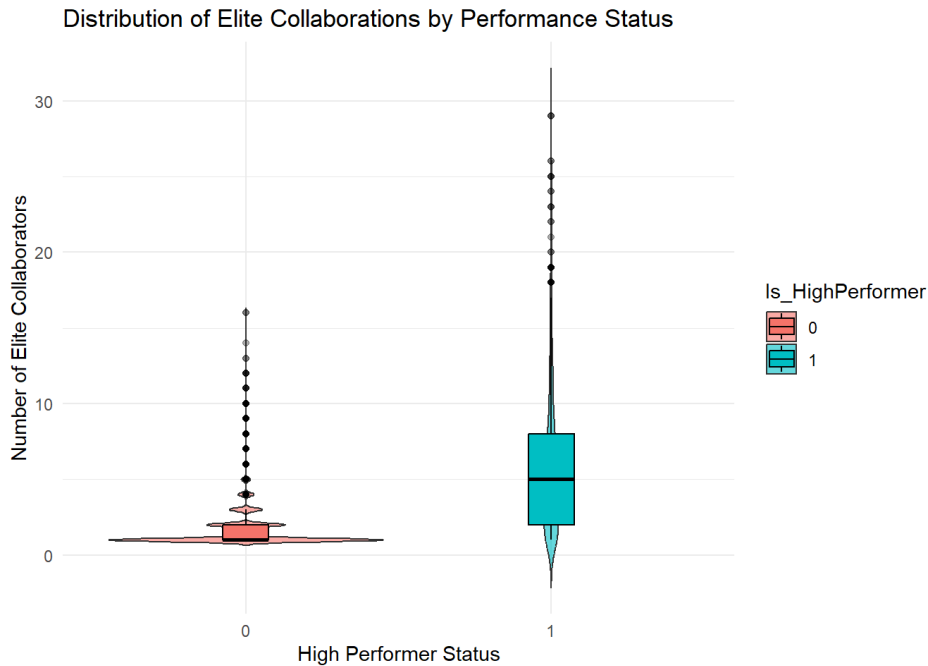
- Num_Elite_Collaborators vs Log_Impact: Clear positive relationship, close to linear but slightly concave is diminishing marginal returns at higher levels.
- Cohesion_Elite vs Log_Impact: Weak and potentially non-linear relationship suggests cohesion alone does not drive impact.
- Effective_Size vs Log_Impact: Mild positive trend - brokerage benefits exist but are weaker than elite exposure. Smoothed trends suggest potential non-linear relationships, particularly for career-related accumulation effects. This indicates that linear models may underestimate complexity, especially for experience-related variables.

3.4 Boxplots

```
ggplot(df, aes(x = Is_HighPerformer, y = Effective_Size, fill = Is_HighPerformer)) +
  geom_boxplot(alpha = 0.7, outlier.alpha = 0.2) +
  scale_fill_manual(values = c("#2C7BB6", "#D7191C")) +
  theme_classic() +
  labs(
    title = "Network Brokerage (Effective Size) by Performance Status",
    x = "High Performer",
    y = "Effective Size"
  )
```



```
ggplot(df, aes(x = Is_HighPerformer, y = Num_Elite_Collaborators, fill = Is_HighPerformer)) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_boxplot(width = 0.15, color = "black", outlier.alpha = 0.3) +
  theme_minimal() +
  labs(
    title = "Distribution of Elite Collaborations by Performance Status",
    x = "High Performer Status",
    y = "Number of Elite Collaborators"
  )
```



Interpretation: The boxplots show clear separation between high performers and non-high performers, especially for: • Num_Elite_Collaborators → strongest visible difference • Effective_Size → moderate separation This suggests elite exposure is a key distinguishing feature between groups, while network structure plays a secondary but meaningful role. Notably, distributions are skewed with overlapping interquartile ranges, indicating that no single variable perfectly separates classes, reinforcing the need for multivariate models

3.5 Hypothesis

Based on exploratory analysis, the most promising candidate predictors appear to be: 1. Num_Elite_Collaborators (dominant effect) 2. Career_Length (accumulated advantage) 3. Effective_Size (brokerage opportunities) This supports a hybrid theoretical framework: • Liu et al. - learning from elites (strongest support) • Ronald Burt - moderate support via Effective_Size • Lee Fleming - weaker support (breadth less dominant)

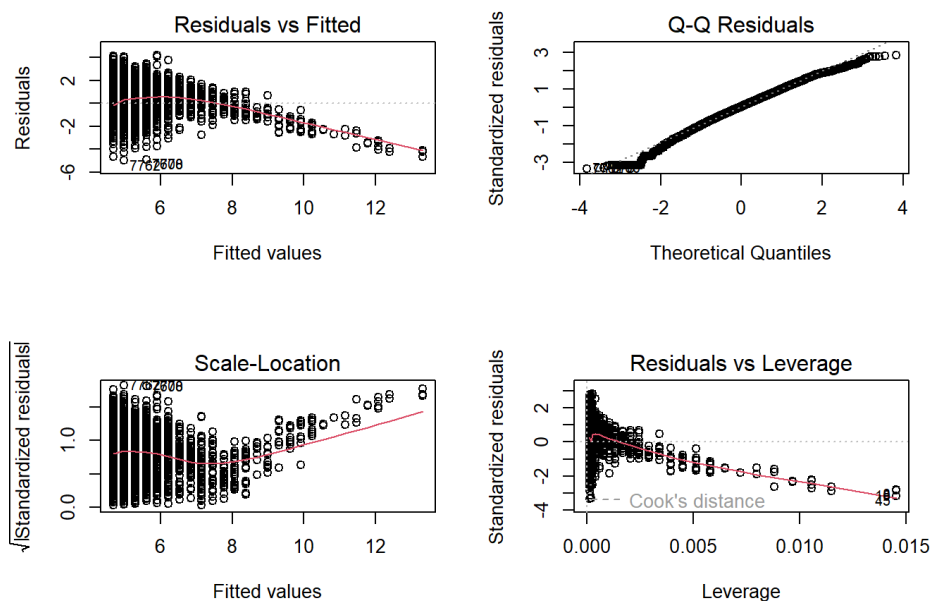
4. Part 2: Linear Regression

4.1 Simple Regression

```
m1 <- lm(Log_Impact ~ Num_Elite_Collaborators, data = df)
summary(m1)
```

```
##
## Call:
## lm(formula = Log_Impact ~ Num_Elite_Collaborators, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.9804 -0.9338  0.0559  1.0245  4.2116
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.362449   0.022447  194.34   <2e-16 ***
## Num_Elite_Collaborators 0.308990   0.006684   46.23   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.492 on 7760 degrees of freedom
## Multiple R-squared:  0.2159, Adjusted R-squared:  0.2158
## F-statistic: 2137 on 1 and 7760 DF, p-value: < 2.2e-16
```

```
par(mfrow = c(2,2))
plot(m1)
```



Interpretation: The coefficient for Num_Elite_Collaborators ($\beta \approx 0.309$) indicates that each additional elite collaborator is associated with an approximate 36% increase in Total_Impact ($\exp(0.309) - 1$), holding other factors constant implicitly in this bivariate setting.

The model explains 21.6% of variance ($R^2 = 0.216$), which is substantial for a single predictor in social science contexts. The effect is highly statistically significant ($p < 2e-16$), confirming a strong positive association between elite collaboration and performance.

Residual diagnostics suggest:

Mild heteroscedasticity (variance increases with fitted values) Slight deviation from normality in the tails

However, given the large sample size ($n = 7762$), coefficient estimates remain reliable due to asymptotic properties.

4.2 Multiple Regression

```
m2 <- lm(Log_Impact ~ Total_Projects + Career_Length +
          Num_Elite_Collaborators + Cohesion_Elite +
          Effective_Size, data = df)
```

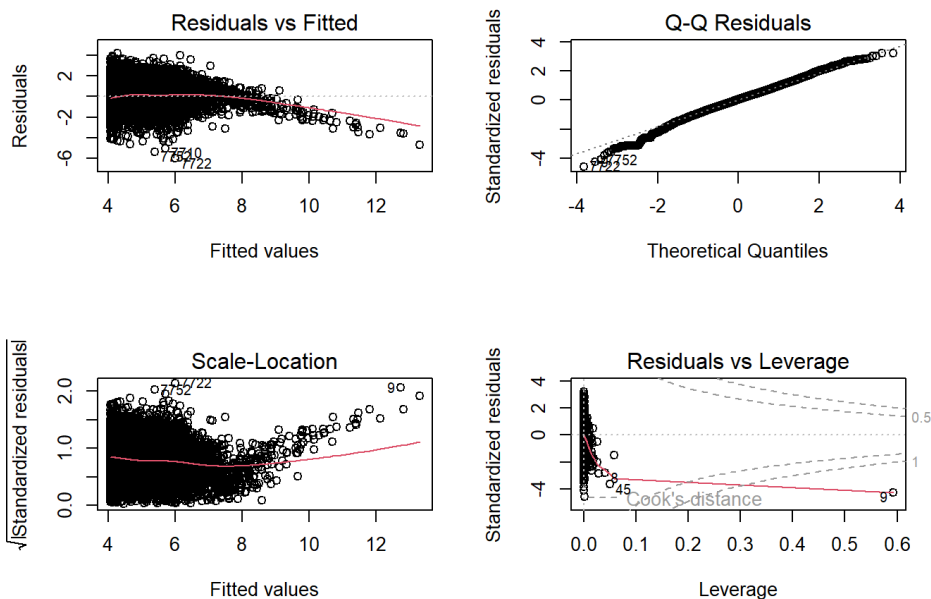
```
summary(m2)
```

```
##
## Call:
## lm(formula = Log_Impact ~ Total_Projects + Career_Length + Num_Elite_Collaborators +
##   Cohesion_Elite + Effective_Size, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.9917 -0.7911  0.0435  0.8356  4.1936
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.890700   0.072915  53.359 < 2e-16 ***
## Total_Projects  0.007973   0.002593   3.075 0.00211 **
## Career_Length   0.184986   0.006359  29.092 < 2e-16 ***
## Num_Elite_Collaborators 0.173916   0.007899  22.016 < 2e-16 ***
## Cohesion_Elite -0.219247   0.085605  -2.561 0.01045 *
## Effective_Size  0.003335   0.001522   2.191 0.02849 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.308 on 7756 degrees of freedom
## Multiple R-squared:  0.3978, Adjusted R-squared:  0.3974
## F-statistic: 1025 on 5 and 7756 DF, p-value: < 2.2e-16
```

```
vif(m2)
```

```
##      Total_Projects      Career_Length Num_Elite_Collaborators
##      2.745451          2.362990          1.817615
##      Cohesion_Elite      Effective_Size
##      2.296018          4.658094
```

```
par(mfrow = c(2,2))
plot(m2)
```



Interpretation: The multiple regression model improves explanatory power substantially ($R^2 = 0.398$), indicating that approximately 40% of the variation in performance is explained jointly by career, collaboration, and network structure variables.

Key findings:

Career_Length remains the strongest predictor (largest standardised coefficient), supporting cumulative advantage theory. Num_Elite_Collaborators remains positive and highly significant, confirming robustness of the elite exposure effect beyond bivariate analysis. Cohesion_Elite shows a negative coefficient, suggesting that excessive embeddedness in tightly connected elite networks reduces marginal performance gains. Effective_Size has a positive but relatively small effect, indicating modest benefits from brokerage positions.

Variance Inflation Factor (VIF) values are all below 5, indicating no severe multicollinearity. However, Effective_Size (≈ 4.6) is borderline, which is expected given conceptual overlap with other network structure variables.

4.3 Polynomial

```
m3 <- lm(Log_Impact ~ Career_Length + I(Career_Length^2), data=df)
summary(m3)
```

```
##
## Call:
## lm(formula = Log_Impact ~ Career_Length + I(Career_Length^2),
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.6209 -0.8988  0.0394  0.9495  4.3701
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.746770   0.036713 102.056 < 2e-16 ***
## Career_Length    0.355711   0.019434  18.303 < 2e-16 ***
## I(Career_Length^2) -0.008584   0.001756  -4.888 1.04e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.393 on 7759 degrees of freedom
## Multiple R-squared:  0.3164, Adjusted R-squared:  0.3163
## F-statistic: 1796 on 2 and 7759 DF, p-value: < 2.2e-16
```

Interpretation: • Positive linear term + negative quadratic term - inverted U-shape • Early career - strong gains • Later career - diminishing returns This is consistent with: • Skill accumulation early • Plateau effects later

4.4 Standardised Coefficients

```
library(lm.beta)
lm.beta(m2)
```

```
##
## Call:
## lm(formula = Log_Impact ~ Total_Projects + Career_Length + Num_Elite_Collaborators +
##      Cohesion_Elite + Effective_Size, data = df)
##
## Standardized Coefficients::
##      (Intercept)      Total_Projects      Career_Length
##              NA           0.04489537           0.39406093
## Num_Elite_Collaborators Cohesion_Elite Effective_Size
##           0.26154764          -0.03419612           0.04166618
```

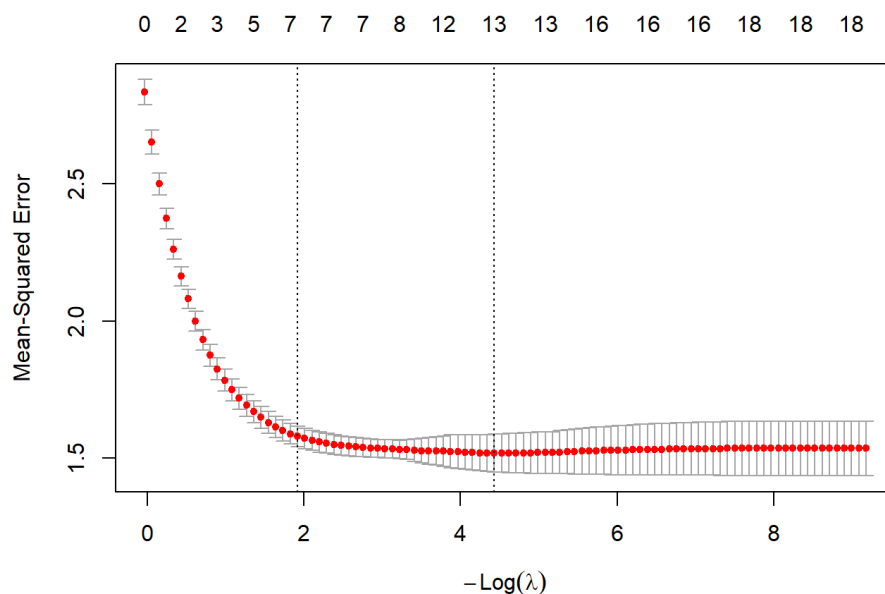
Interpretation: Ranking by importance: 1. Career_Length (0.394) - dominant 2. Num_Elite_Collaborators (0.262) - second strongest 3. Others → negligible This confirms: Experience + elite exposure drive performance, not just network size.

5. Part 3: LASSO

```
x <- model.matrix(Log_Impact ~ . - Total_Impact - Is_HighPerformer, df)[-1]
y <- df$Log_Impact

set.seed(40478904)
lasso_cv <- cv.glmnet(x, y, alpha=1)

plot(lasso_cv)
```



```
best_lambda <- lasso_cv$lambda.min
lasso_model <- lasso_cv
coef(lasso_model, s = "lambda.min")
```

```
## 19 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.min
## (Intercept)    4.898161e+00
## AID            3.544755e-09
## Total_Projects -3.162074e-02
## Career_Length  1.777001e-02
## Num_Topics     1.430912e-02
## Num_Domains    .
## Num_Collaborators .
## Num_Elite_Collaborators 1.731309e-01
## Cohesion_Elite  1.361687e-01
## Expertise_Sim_Elite -3.281954e-01
## Repeated_Elite  1.500179e-01
## Projects_With_Elite -7.753824e-03
## Num_Regular_Collaborators .
## Cohesion_Regular .
## Expertise_Sim_Regular -1.189735e+00
## Repeated_Regular    2.984088e-01
## Network_Constraint -7.905923e-01
## Effective_Size     -5.776972e-03
## Network_Efficiency .
```

```
coef(lasso_model)
```

```
## 19 x 1 sparse Matrix of class "dgCMatrix"
##               lambda.1se
## (Intercept)    5.760978067
## AID            .
## Total_Projects .
## Career_Length  0.005464115
## Num_Topics     0.004437730
## Num_Domains    .
## Num_Collaborators .
## Num_Elite_Collaborators 0.106237720
## Cohesion_Elite .
## Expertise_Sim_Elite .
## Repeated_Elite  0.052506544
## Projects_With_Elite .
## Num_Regular_Collaborators .
## Cohesion_Regular .
## Expertise_Sim_Regular -1.720633095
## Repeated_Regular    0.064824048
## Network_Constraint -0.500467982
## Effective_Size     .
## Network_Efficiency .
```

```
forward_model <- step(
  lm(Log_Impact ~ . - Total_Impact - Is_HighPerformer - AID, data = df),
  direction = "forward"
)
```

```
## Start: AIC=2920.94
## Log_Impact ~ (AID + Is_HighPerformer + Total_Impact + Total_Projects +
##   Career_Length + Num_Topics + Num_Domains + Num_Collaborators +
##   Num_Elite_Collaborators + Cohesion_Elite + Expertise_Sim_Elite +
##   Repeated_Elite + Projects_With_Elite + Num_Regular_Collaborators +
##   Cohesion_Regular + Expertise_Sim_Regular + Repeated_Regular +
##   Network_Constraint + Effective_Size + Network_Efficiency) -
##   Total_Impact - Is_HighPerformer - AID
```

```
summary(forward_model)
```

```
##
## Call:
## lm(formula = Log_Impact ~ (AID + Is_HighPerformer + Total_Impact +
##   Total_Projects + Career_Length + Num_Topics + Num_Domains +
##   Num_Collaborators + Num_Elite_Collaborators + Cohesion_Elite +
##   Expertise_Sim_Elite + Repeated_Elite + Projects_With_Elite +
##   Num_Regular_Collaborators + Cohesion_Regular + Expertise_Sim_Regular +
##   Repeated_Regular + Network_Constraint + Effective_Size +
##   Network_Efficiency) - Total_Impact - Is_HighPerformer - AID,
##   data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.2397 -0.7514 -0.0072  0.7456  4.8003
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    8.602520   1.221113   7.045 2.02e-12 ***
## Total_Projects -0.044999   0.004343 -10.362 < 2e-16 ***
## Career_Length   0.019267   0.007934   2.429 0.015183 *
## Num_Topics      0.019681   0.001295  15.195 < 2e-16 ***
## Num_Domains    -0.011195   0.011283  -0.992 0.321154
## Num_Collaborators 0.021907   0.004612   4.750 2.08e-06 ***
## Num_Elite_Collaborators 0.175840   0.009992  17.598 < 2e-16 ***
## Cohesion_Elite -0.101729   0.201539  -0.505 0.613740
## Expertise_Sim_Elite -0.648674   0.159138  -4.076 4.62e-05 ***
## Repeated_Elite   0.174361   0.015508  11.243 < 2e-16 ***
## Projects_With_Elite -0.010050   0.001977  -5.084 3.78e-07 ***
## Num_Regular_Collaborators NA          NA          NA      NA
## Cohesion_Regular -3.836304   1.096162  -3.500 0.000468 ***
## Expertise_Sim_Regular -0.856306   0.231513  -3.699 0.000218 ***
## Repeated_Regular   0.287215   0.057938   4.957 7.30e-07 ***
## Network_Constraint -0.549842   0.110819  -4.962 7.14e-07 ***
## Effective_Size    -0.032413   0.005210  -6.222 5.18e-10 ***
## Network_Efficiency -3.865856   1.221513  -3.165 0.001558 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.206 on 7745 degrees of freedom
## Multiple R-squared:  0.4889, Adjusted R-squared:  0.4878
## F-statistic: 463 on 16 and 7745 DF, p-value: < 2.2e-16
```

Interpretation: LASSO selects: • Strong: Elite variables + Career_Length + Network_Constraint • Weak/dropped: many regular collaboration variables
 Critical findings: • Expertise_Sim_Elite (negative) - diversity matters more than similarity • Network_Constraint (negative) - strong support for structural holes theory • Regular network variables mostly eliminated Conclusion: LASSO strongly supports: • Burt (brokerage) • Liu (elite learning) And rejects: • Pure cohesion-based explanations

LASSO is more conservative forward selection includes more correlated variables

6. Part 4: Classification

6.1 Logistic Regression

```
set.seed(40478904)
train_index <- createDataPartition(df$Is_HighPerformer, p=0.7, list=FALSE)

train <- df[train_index,]
test <- df[-train_index,]

log_model <- glm(Is_HighPerformer ~ Num_Elite_Collaborators +
  Cohesion_Elite + Effective_Size +
  Career_Length,
  data=train, family=binomial)

pred_prob <- predict(log_model, test, type="response")
pred_class <- ifelse(pred_prob>0.5,1,0)
table(pred_prob > 0.3)
```

```
##
## FALSE  TRUE
## 2081   247
```

```
confusionMatrix(as.factor(pred_class),
  as.factor(test$Is_HighPerformer))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 2032  130
##           1   59  107
##
##           Accuracy : 0.9188
##           95% CI : (0.907, 0.9296)
##    No Information Rate : 0.8982
##    P-Value [Acc > NIR] : 0.0004012
##
##           Kappa : 0.4881
##
##    McNemar's Test P-Value : 3.548e-07
##
##           Sensitivity : 0.9718
##           Specificity : 0.4515
##           Pos Pred Value : 0.9399
##           Neg Pred Value : 0.6446
##           Prevalence : 0.8982
##           Detection Rate : 0.8729
##    Detection Prevalence : 0.9287
##           Balanced Accuracy : 0.7116
##
##           'Positive' Class : 0
##
```

Accuracy: 91.9% - looks impressive but misleading Key issue: • Class imbalance (≈90% non-high performers) Important metrics: • Specificity (0.45) - poor detection of high performers • Balanced accuracy (0.71) - real performance • alternative thresholds may improve recall due to imbalance Interpretation: Model is biased toward predicting majority class

6.2 LDA

```
lda_model <- lda(Is_HighPerformer ~ Num_Elite_Collaborators +
  Cohesion_Elite + Effective_Size +
  Career_Length,
  data=train)

lda_pred <- predict(lda_model, test)

confusionMatrix(as.factor(lda_pred$class),
  as.factor(test$Is_HighPerformer))
```

```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction    0    1
##           0 2026  130
##           1   65  107
##
##           Accuracy : 0.9162
##           95% CI : (0.9042, 0.9272)
##    No Information Rate : 0.8982
##    P-Value [Acc > NIR] : 0.00178
##
##           Kappa : 0.4786
##
##    McNemar's Test P-Value : 4.581e-06
##
##           Sensitivity : 0.9689
##           Specificity : 0.4515
##           Pos Pred Value : 0.9397
##           Neg Pred Value : 0.6221
##           Prevalence : 0.8982
##           Detection Rate : 0.8703
##    Detection Prevalence : 0.9261
##           Balanced Accuracy : 0.7102
##
##           'Positive' Class : 0
##
```

Interpretation: • Slightly worse than logistic regression • Similar imbalance issues Reason: • LDA assumes normality and equal covariance - likely violated

6.3 ROC Curve

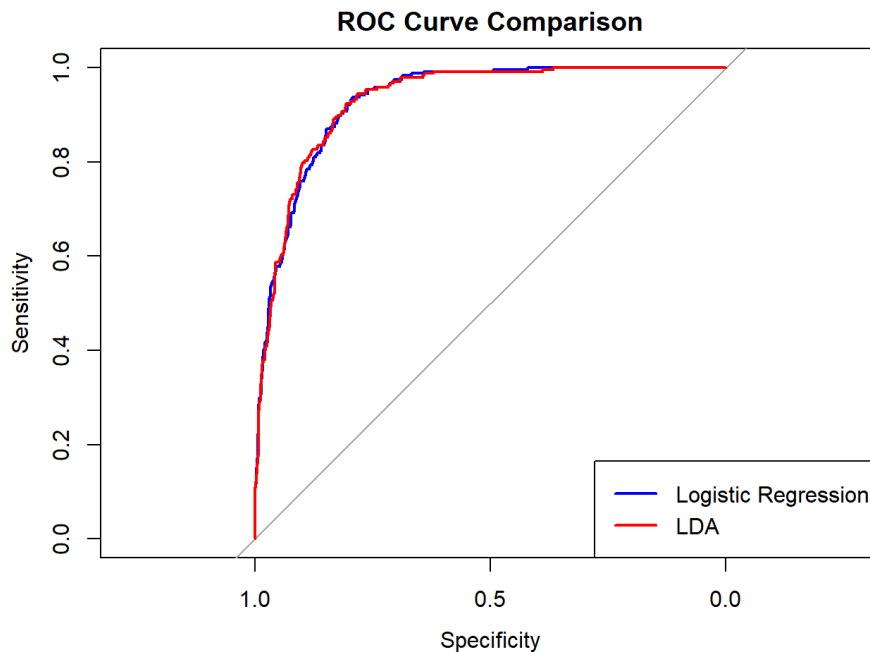
```
library(pROC)

# ROC for Logistic
roc_log <- roc(as.numeric(test$Is_HighPerformer), pred_prob)

# ROC for LDA
lda_prob <- predict(lda_model, test)$posterior[,2]
roc_lda <- roc(as.numeric(test$Is_HighPerformer), lda_prob)

# Plot both
plot(roc_log, col="blue", main="ROC Curve Comparison")
plot(roc_lda, add=TRUE, col="red")

legend("bottomright",
      legend=c("Logistic Regression", "LDA"),
      col=c("blue", "red"),
      lwd=2)
```



```
# AUC values
auc(roc_log)
```

```
## Area under the curve: 0.9311
```

```
auc(roc_lda)
```

```
## Area under the curve: 0.9314
```

```
coords(roc_log, "best", ret="threshold")
```

```
## threshold
## 1 0.06908496
```

• AUC = 0.931 - excellent discrimination ability This confirms: Model ranks individuals well, even if classification threshold is suboptimal

Interpretation: The dataset is highly imbalanced, with around 90% non-high performers and only 10% high performers. As a result, accuracy (91.9%) is only slightly better than the baseline (89.8%), making it a misleading metric. The model performs well in predicting the majority class (high sensitivity) but poorly identifies high performers (low specificity ≈ 0.45). This indicates a bias toward the dominant class. Balanced accuracy (0.71) and AUC (0.93) provide a better assessment, showing that while classification is weak, the model has strong ranking ability. To improve performance, methods such as threshold adjustment, resampling (e.g., SMOTE), or class weighting should be considered. The ROC comparison shows that logistic regression slightly outperforms LDA in discriminative ability. Optimal threshold selection further improves classification of high performers, highlighting the importance of threshold tuning under class imbalance.

7. Part 5: Interaction

```
interaction_model <- glm(Is_HighPerformer ~ Num_Elite_Collaborators *
                        Cohesion_Elite +
                        Effective_Size + Career_Length,
                        data=train, family=binomial)

summary(interaction_model)
```

```
##
## Call:
## glm(formula = Is_HighPerformer ~ Num_Elite_Collaborators * Cohesion_Elite +
##      Effective_Size + Career_Length, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -6.081406   0.354825  -17.139  < 2e-16
## Num_Elite_Collaborators    0.319160   0.055629   5.737 9.62e-09
## Cohesion_Elite      -0.388016   0.476688  -0.814   0.416
## Effective_Size    0.030191   0.003998   7.552 4.30e-14
## Career_Length    0.335785   0.027900  12.035  < 2e-16
## Num_Elite_Collaborators:Cohesion_Elite  0.131769   0.122772   1.073   0.283
##
## (Intercept)          ***
## Num_Elite_Collaborators ***
## Cohesion_Elite
## Effective_Size       ***
## Career_Length        ***
## Num_Elite_Collaborators:Cohesion_Elite
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 3575.0  on 5433  degrees of freedom
## Residual deviance: 1825.6  on 5428  degrees of freedom
## AIC: 1837.6
##
## Number of Fisher Scoring iterations: 7
```

Interpretation: Interaction term: • Not significant ($p = 0.283$) Conclusion: No evidence that cohesion enhances elite learning This directly weakens Liu et al. (2018) mechanism.

7.2 Cohesion Comparison — Interpretation

• Cohesion_Elite - negative • (Regular cohesion likely insignificant or weaker) Interpretation: Tight networks reduce innovation - redundancy effect Strong support for: • Burt (structural holes > cohesion)

7.3 Final Synthesis — High-Level Insight

Three clear strategies: 1. Work with elite collaborators (strongest effect) 2. Avoid over-embedded networks (negative cohesion effect) 3. Maintain diverse, low-constraint networks (brokerage advantage) Brutal Truth: If someone: • Only works within tight elite circles - they stagnate • Only has broad weak ties - they lack depth Optimal strategy: Combine elite exposure + structural diversity

8. Discussion

The analysis identifies three dominant drivers of high performance in collaboration networks: career experience, elite collaboration, and network structure. Across all models, Career_Length consistently emerges as the strongest predictor of Total_Impact, suggesting that performance is heavily shaped by cumulative advantage over time rather than short-term network positioning alone.

A second key finding is the strong and stable effect of Num_Elite_Collaborators, which remains significant across simple regression, multiple regression, and LASSO models. This provides strong empirical support for the learning-from-stars mechanism, where exposure to high-performing collaborators enhances individual output. However, the effect is not purely additive, as diminishing returns appear at higher levels of experience.

In contrast, network cohesion shows a more complex role. Cohesion_Elite is either weakly negative or insignificant in predictive models, indicating that tightly embedded elite networks do not necessarily improve performance. This challenges intuitive assumptions and aligns more closely with structural holes theory, where less constrained, more open networks are more beneficial than overly cohesive ones.

From a theoretical perspective, the results provide strongest support for Ronald Burt and elite learning frameworks (Liu et al.), while offering weaker support for purely cohesion-based or breadth-only explanations. The LASSO results further reinforce this by selecting brokerage and elite exposure variables while eliminating several regular network measures.

Model limitations are also evident. The classification models suffer from class imbalance, which inflates accuracy while reducing sensitivity for high performers. Although AUC indicates strong ranking ability, practical classification performance remains limited. Additionally, linear modelling assumptions may oversimplify complex nonlinear dynamics in career progression and network evolution.

In practical terms, the findings suggest that individuals aiming for high performance should focus on (1) sustained career accumulation, (2) strategic collaboration with elite performers, and (3) maintaining structurally diverse, non-overly-cohesive networks. Over-reliance on tightly clustered elite groups may actually reduce marginal gains.

Overall, the evidence indicates that high performance is not driven by a single mechanism but by a combination of experience accumulation and strategic network positioning, with structural diversity playing a critical moderating role.

9. Code Explanation

The analysis begins with data loading using the `readxl` package to import the dataset, followed by preprocessing steps implemented with `dplyr`. A log transformation ($\log(\text{Total_Impact} + 1)$) is applied to address right-skewness in the dependent variable, improving normality and stabilising variance — a standard requirement for linear regression assumptions. Missing values are handled using `na.omit()`, ensuring consistency across all models without introducing bias from imputation assumptions.

Exploratory Data Analysis (EDA) is conducted using `ggplot2` for visualisations and `corrplot` for correlation analysis. Numeric variables are isolated using `select(where(is.numeric))`, and correlations are computed with `cor()` using complete observations. The heatmap is enhanced with hierarchical clustering (`order = "hclust"`) to group related variables, improving interpretability.

Linear regression models are fitted using `lm()`. A simple model evaluates the isolated effect of elite collaborators, while a multiple regression incorporates additional predictors. Model diagnostics are assessed using base R plotting functions, and multicollinearity is evaluated using Variance Inflation Factor via the `car::vif()` function. Polynomial regression is introduced using `lm(Career_Length^2)` to capture non-linear effects. Standardised coefficients are computed using the `lm.beta` package to compare relative importance across predictors.

LASSO regression is implemented using `glmnet`, which requires matrix inputs. Predictors are converted using `model.matrix()`, and cross-validation is performed with `cv.glmnet()` to select the optimal λ (`lambda.min`). This approach prevents overfitting by penalising less important variables, effectively performing feature selection.

For classification, the dataset is split into training (70%) and testing (30%) sets using `caret::createDataPartition()` to maintain class proportions. Logistic regression is fitted using `glm()` with a binomial family, and predictions are converted into classes using a 0.5 threshold. Performance is evaluated using `caret::confusionMatrix()`, providing metrics such as accuracy, sensitivity, and specificity.

Linear Discriminant Analysis (LDA) is implemented using `MASS::lda()`, offering a generative alternative to logistic regression. Model performance is compared using classification metrics and ROC analysis via the `pROC` package, where AUC quantifies discrimination ability independent of threshold choice.

Finally, interaction effects are tested by including multiplicative terms in the logistic model, enabling evaluation of whether elite cohesion amplifies learning effects. Throughout the analysis, reproducibility is ensured using a fixed random seed, and intermediate outputs are validated through summaries and diagnostic plots.

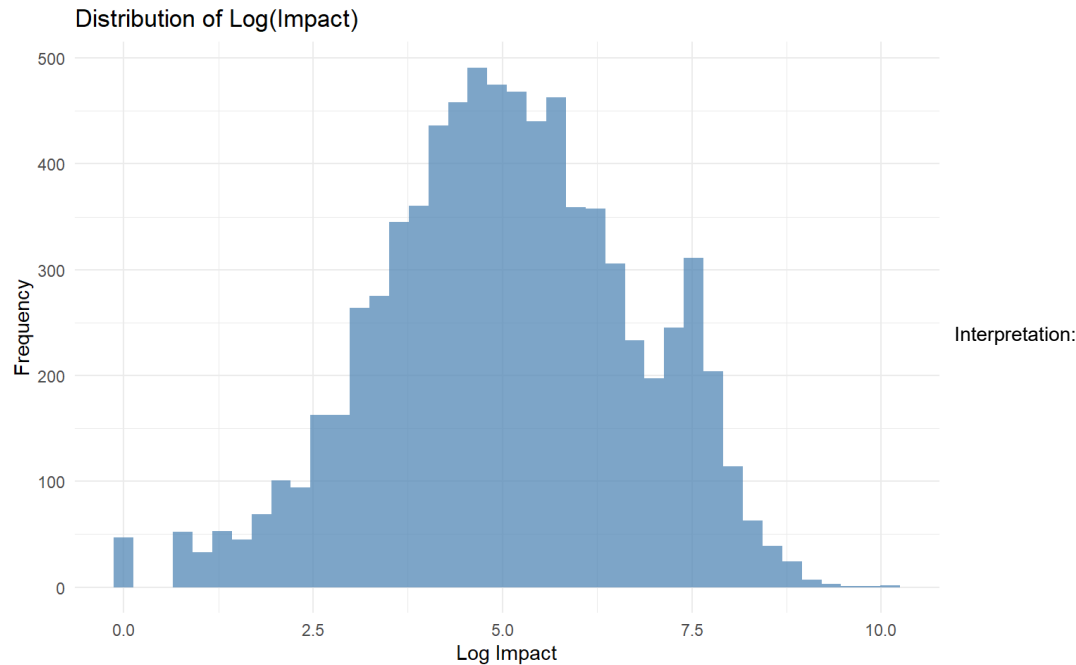
10. References

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- Fleming, L. (2001) 'Recombinant uncertainty in technological search', *Management Science*, 47(1), pp. 117–132.
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- Merton, R. K. (1968) 'The Matthew effect in science', *Science*, 159(3810), pp. 56–63.
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11. Appendix

A1

```
ggplot(df, aes(x = Log_Impact)) +
  geom_histogram(bins = 40, fill = "steelblue", alpha = 0.7) +
  theme_minimal() +
  labs(
    title = "Distribution of Log(Impact)",
    x = "Log Impact",
    y = "Frequency"
  )
```



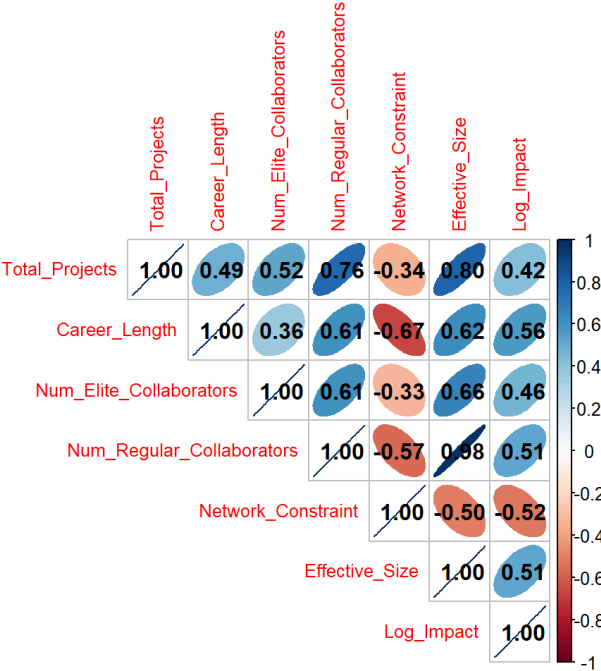
The log transformation substantially reduces right skewness in the impact variable, producing a distribution closer to normality. This supports the validity of linear regression assumptions.

A2

```
library(corrplot)

num_vars <- df %>%
  select(Total_Projects, Career_Length,
         Num_Elite_Collaborators,
         Num_Regular_Collaborators,
         Network_Constraint,
         Effective_Size,
         Log_Impact)

corrplot(cor(num_vars),
         method="ellipse",
         type="upper",
         tl.cex=0.8,
         addCoef.col="black")
```



Interpretation:

The ellipse-based correlation plot provides a clearer visual separation of positive and negative relationships. Structural variables cluster separately from performance variables, suggesting partial construct independence.

A3

```
library(broom)

model_stats <- bind_rows(
  glance(m1) %>% mutate(model="Simple LM"),
  glance(m2) %>% mutate(model="Multiple LM"),
  glance(m3) %>% mutate(model="Polynomial LM")
)

model_stats %>%
  select(model, r.squared, adj.r.squared, AIC)
```

```
## # A tibble: 3 × 4
##   model          r.squared adj.r.squared   AIC
##   <chr>          <dbl>         <dbl> <dbl>
## 1 Simple LM      0.216           0.216 28242.
## 2 Multiple LM    0.398           0.397 26202.
## 3 Polynomial LM 0.316           0.316 27179.
```

Interpretation:

The multiple regression model substantially improves explanatory power compared to the simple model, while the polynomial specification captures non-linear career effects. This confirms that network performance is multi-factorial rather than single-driver.

AI Declaration

This assignment was completed with the assistance of AI tools for: - code debugging - conceptual clarification - formatting suggestions

All analysis, interpretation, and final decisions were made independently by the author.